

Randomized and Quantum Lifting for One-Way Conservative NOF Model

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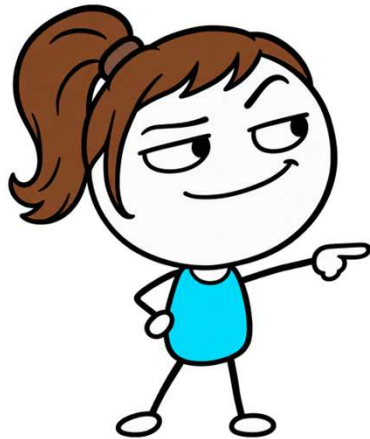


Part 1. Introduction

One-way 2-party model



r : public randomness



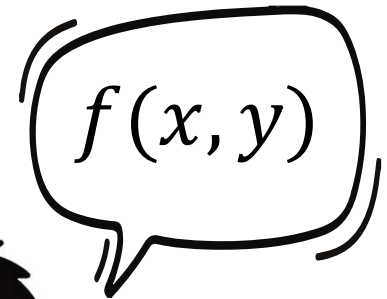
Alice: y



$\Pi(y, r): 1011011\dots$



Bob: x



Communication complexity

- **Cost:** the length of message $|\Pi(y, r)|$;
- **Goal:** outputs $f(x, y)$ correctly with probability $\geq \frac{2}{3}$;

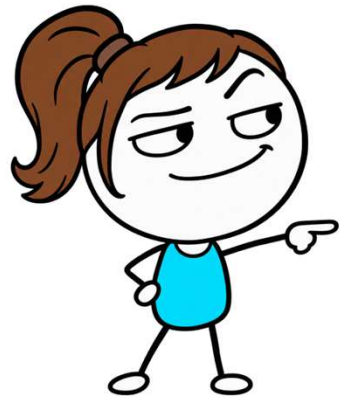
How long must $|\Pi(y, r)|$ be?

$$\text{Let } R_{2/3}^{\rightarrow}(f) := \min_{\Pi} \max_{y, r} |\Pi(y, r)|$$

One-way NOF model



r : public randomness



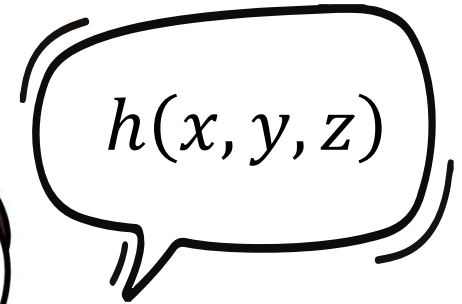
Lingling: y, z



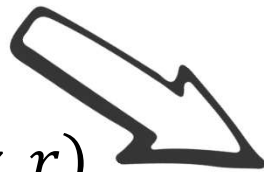
$\Pi_{L,S}(y, z, r)$



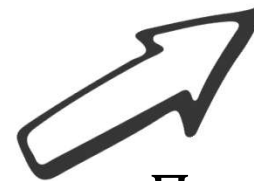
Sam: x, y



$\Pi_{L,D}(y, z, r)$



Daming: x, z



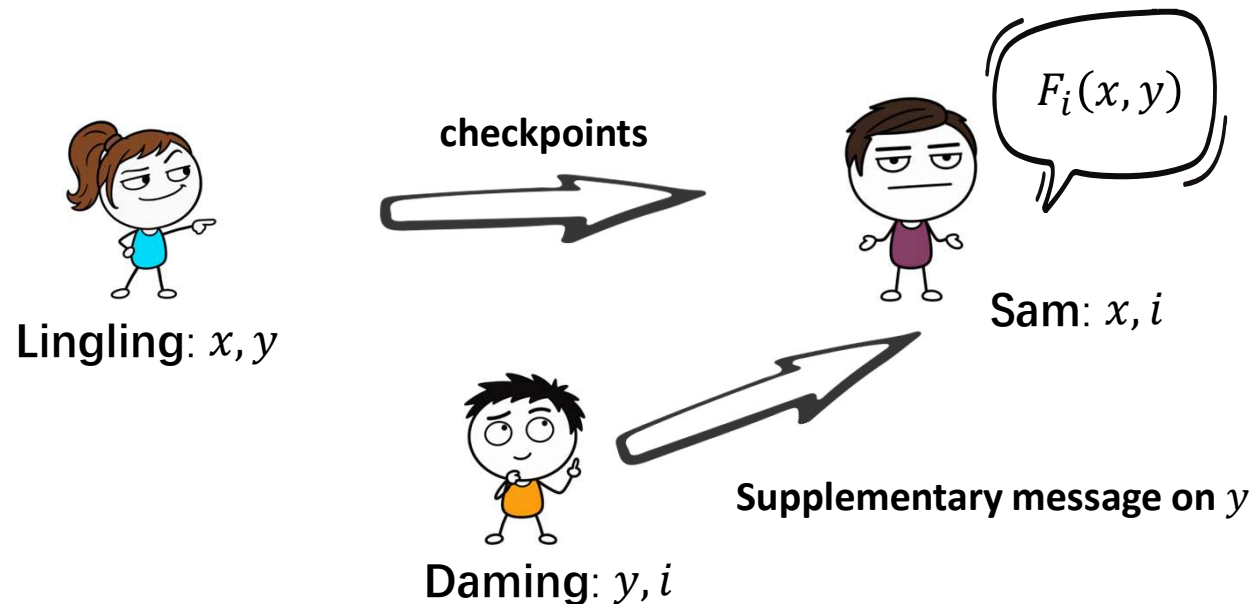
$\Pi_{D,S}(x, z, r)$

Why we study one-way NOF model?

- **Circuit lower bounds:** Small shallow circuits induce low-cost one-way NOF protocols. ----[Håstad–Goldmann, FOCS1990]; [Nisan–Wigderson, SIAM J. Comput.1993]; [Beigel–Tarui, Comput. Complex.1994];
- **Additive combinatorics:** NOF lower bounds are directly tied to progression-free sets, corners, and Ruzsa–Szemerédi graphs. Progress flows both ways. ----[Linial–Pitassi–Shraibman, ITCS2019]; [Linial–Shraibman, CCC2021]; [Jaber et al., FOCS2025];

An example: circuits to protocols

- $F(x, y) = (F_1(x, y), F_2(x, y), \dots, F_n(x, y))$.
- Given (x, y, i) , $i \in [n]$, to compute $F_i(x, y)$.
- [Valiant, MFCS 1977]: if the circuit for $F(x, y)$ is shallow and small, there exists some **key checkpoints**.



Why we study one-way NOF model?

- **Pseudorandomness:** Small-space distinguishers induce low-cost NOF protocols. ----[Babai–Nisan–Szegedy, STOC1989]; [Ganor–Raz, RANDOM2014];
- **Cryptography:** indistinguishability in NOF model powers leakage-resilient cryptography. ----[Kumar–Meka–Sahai, FOCS2019]; [Li–Ma–Quach–Wichs, CRYPTO2020]
- **Other Applications:** Streaming algorithms; Distributed computing; [Kallaughier et al., PODS2019]; [Drucker–Kuhn–Oshman, PODC2014]



Why we study one-way NOF model?

- **Quantum advantage:** randomized one-way NOF lower bounds on certain quantum-friendly function can show quantum advantage;

Open problem: prove strong lower bounds of one-way NOF model.



Research History (one-way NOF, to name a few)

- [Viola–Wigderson, FOCS 2007]: $\text{Poly}(n)$ randomized lower bounds for one-way pointer jumping and DISJ.
- [Jain–Klauck–Nayak, STOC 2008]: $\Omega(\sqrt{n})$ for randomized 3-party one-way DISJ.
- [Gavinsky–Pudlák, CCC 2008]: $n^{\Omega(1/k^2)}$ randomized lower bound for an one-way NOF search problem; exponential quantum advantage.
- **[Yang–Zhang, preprint 2025]: A generic **deterministic lifting** theorem.**
- **[Yang–Zhang, FOCS 2026]: A **lifted** search problem requires $\Omega(n^{1/3}/2^k)$ randomized lower bound; exponential quantum advantage.**

Part 2. Randomized lifting theorem

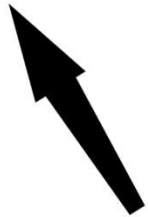
Why randomized NOF lower bound is hard?

- Players' views are highly correlated.
- Hitting (deterministic) vs. Mixing (randomized).
- Generic discrepancy/norm methods lower bound **quantum** and **randomized** models **simultaneously**. ----[Lee–Schechtman–Shraibman, CCC 2009]; [Sherstov, STOC 2013];

One-way two-party model

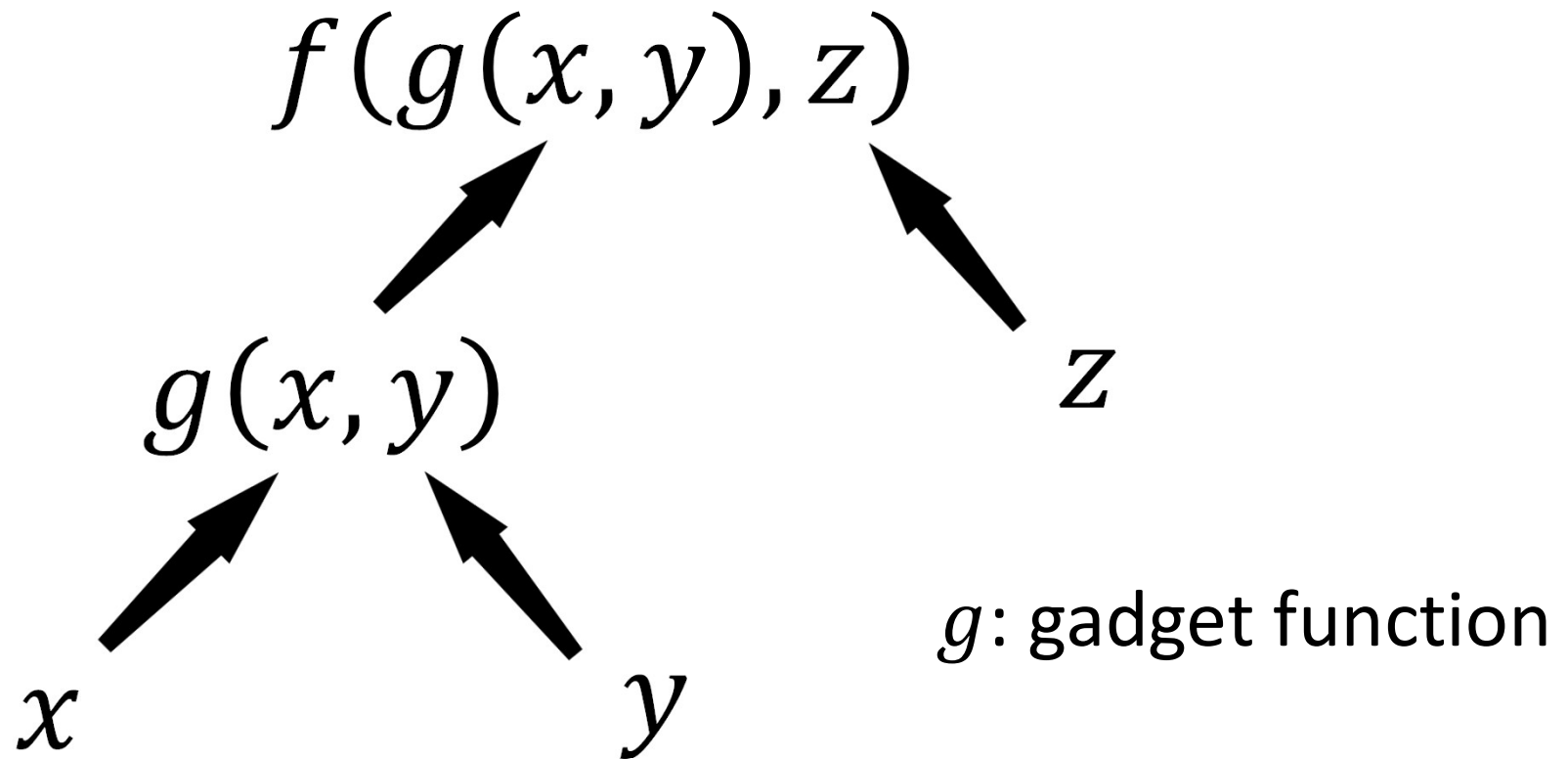


Lifting: transfer lower bounds from a **simple model** to a **richer one**.



One-way NOF model

Lifted function: from 2-party f to 3-party $f \circ g$

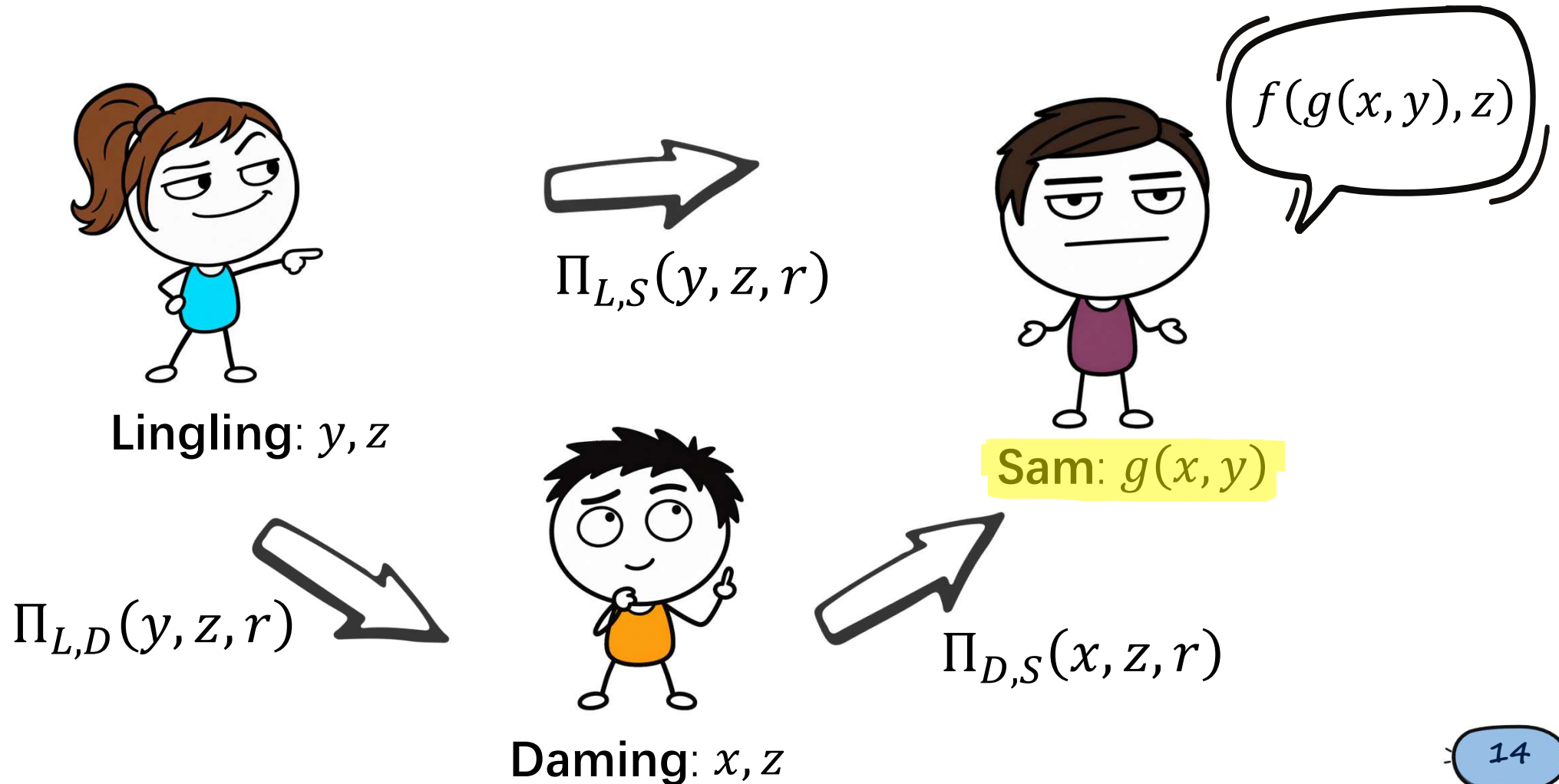


A moderate restriction: **terminal-conservative**

The final player's view:

$g(x, y)$ vs. x, y

One-way conservative NOF model

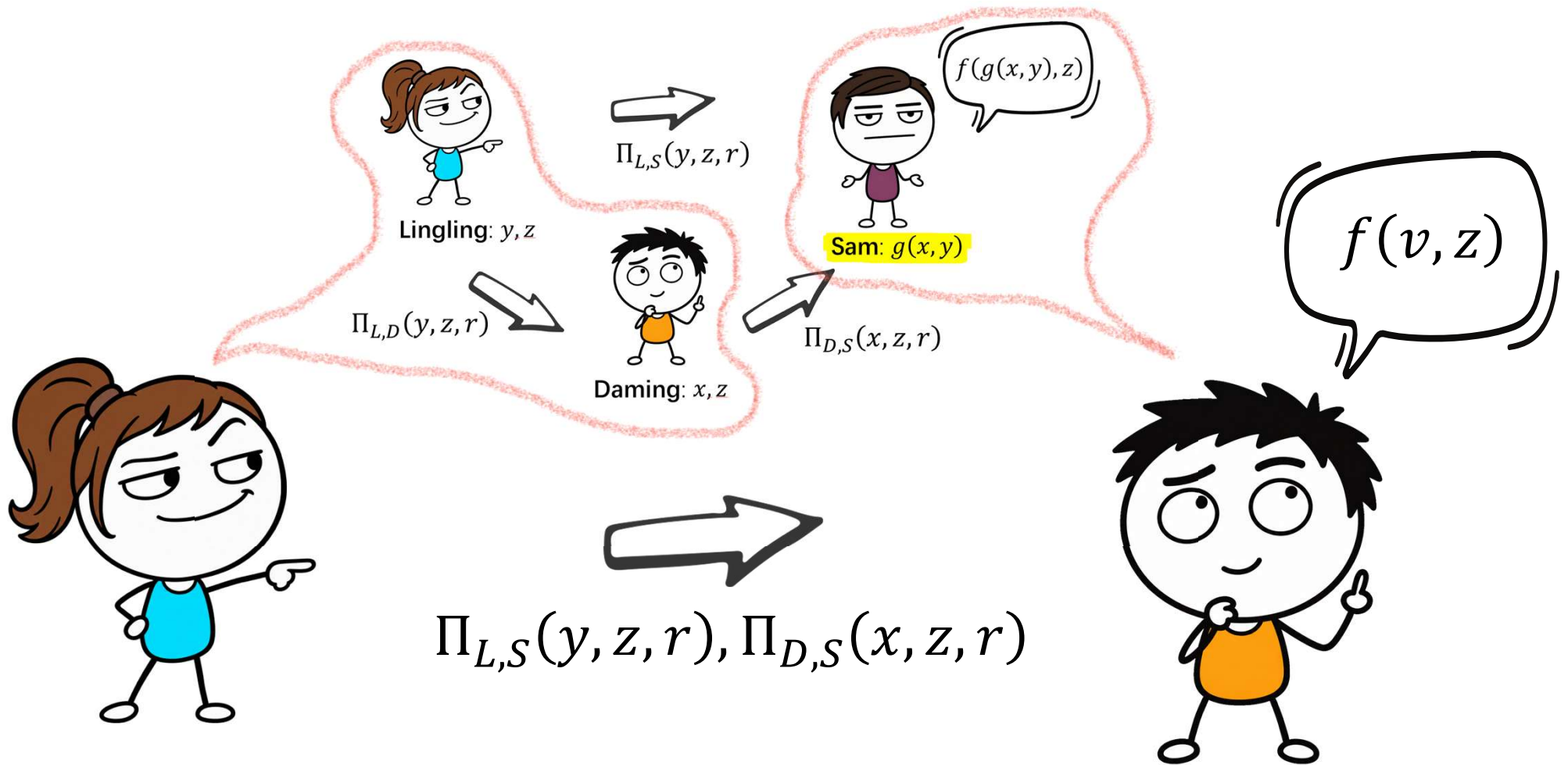


Inner Product: $\text{IP}_{q,r}(x, y) = \sum_{i=1}^r x_i y_i \pmod q$

Let $f \circ \text{IP}_{q,r}(x, y, z) := f(\text{IP}_{q,r}(x, y), z)$

Our result: $\forall f, R_{2/3}^{\rightarrow, \text{cons}}(f \circ \text{IP}_{q,r}) \geq \Omega\left(R_{2/3}^{\rightarrow}(f)\right)$

Idea: each one-way conservative NOF protocol induces a one-way 2-party protocol;



Alice: z ;
sample x, y uniformly;

Bob: v ; takes $g(x, y)$ as v , and outputs
the same answer as **Sam**;

Proof Sketch:

- True transcript $\Pi_{z,r}^{true}(x, y)$: (x, y) is uniform random with $g(x, y) = v$;
- Alice's simulation $\Pi_{z,r}^{sim}(x, y)$: (x, y) is uniform random in the whole universe;
- The pseudo-randomness of $g := \text{IP}_{q,r}$ suggests $\Pi_{z,r}^{true}(x, y) \approx \Pi_{z,r}^{sim}(x, y)$;
* Only high-level intuition; proving such “ $\approx$ ” is the technical centerpiece*
- By the similar transcript and the same decoder, Bob only suffers a small loss on successful probability.

Corollary (exponential quantum advantage):
There exists a **Boolean function** F such that

$$R_{2/3}^{\rightarrow, \text{cons}}(F) = \Omega(\sqrt{n})$$

vs.

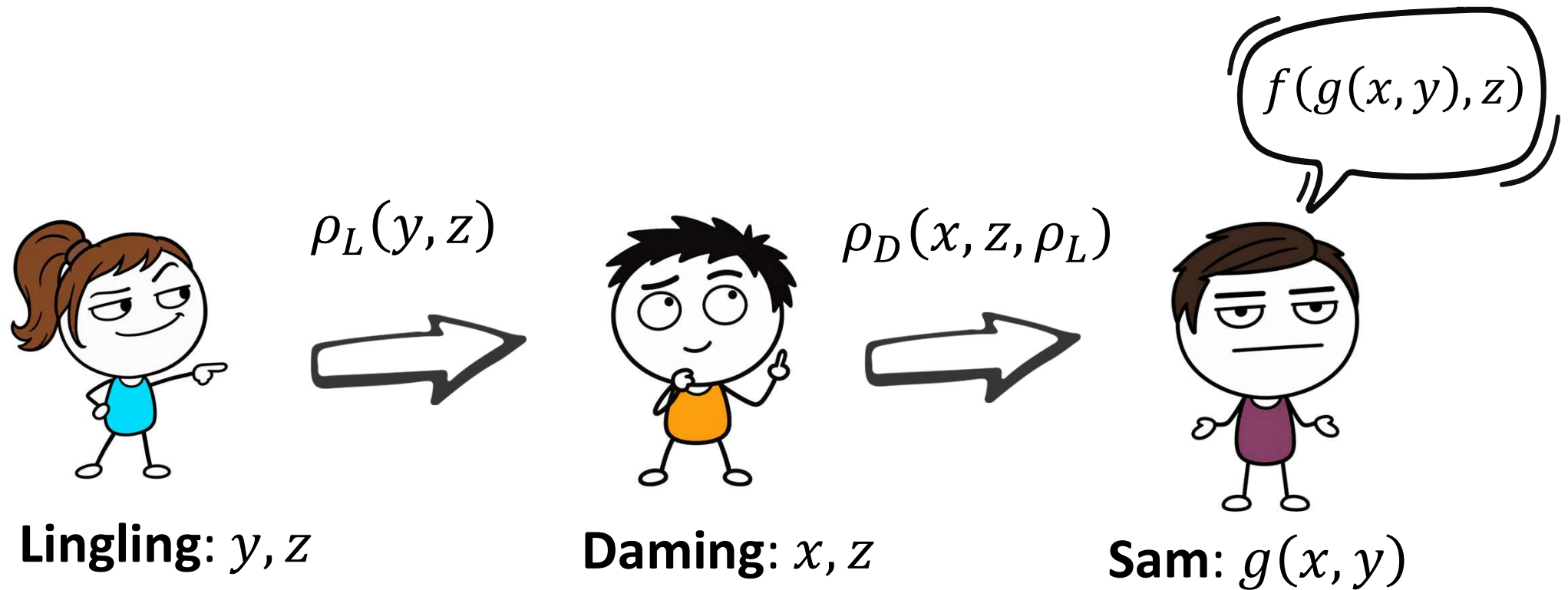
$$Q_{2/3}^{\rightarrow, \text{cons}}(F) = O(\log n)$$

Quantum advantage (to name a few)

Reference	Model	Problem type
[Bar-Yossef–Jayram–Kerenidis, STOC 2004]	One-way 2-party	Search problem
[Gavinsky et al., STOC 2007]	One-way 2-party	Boolean function
[Gavinsky–Pudlák, CCC 2008]	One-way non-interactive NOF	Search problem
[Yang–Zhang, FOCS 2026]	One-way NOF	Search problem
Ours	One-way conservative NOF	Boolean function

Part 3. Quantum lifting theorem.

Quantum one-way conservative NOF model



Sam measures the state $\rho_D(x, z, \rho_L)$ and outputs the answer.

Our result: $\forall f, Q_{2/3}^{\text{cons}, \rightarrow}(f \circ \text{IP}_{q,r}) \geq \Omega(Q_{2/3}^{\rightarrow}(f))$

- Follows by the same protocol simulation idea;
- **Quantum upgrade:** the gadget $\text{IP}_{q,r}$ fools the entire final quantum state—not just a classical transcript—so conditioning on $g(x, y) = v$ barely changes Sam's measurement.
- New toolkit for quantum communication complexity;

Open questions

- Build randomized lifting for the **one-way NOF model**. (originally raised by [Yang–Zhang, preprint 2025])
- **Better explicit gadgets**: improve the dependency on k (number of parties) from $1/2^k$ (Inner product, ours) to $1/\text{poly}(k)$ (random gadgets).
- **Entanglement-assisted model**: can we build analogous quantum lifting theorem when players share entanglement.
- **Interactive/multi-pass** lifting theorems for more general NOF model.

Merci de votre attention !

Thanks for your attention!